Priyanka Yadav

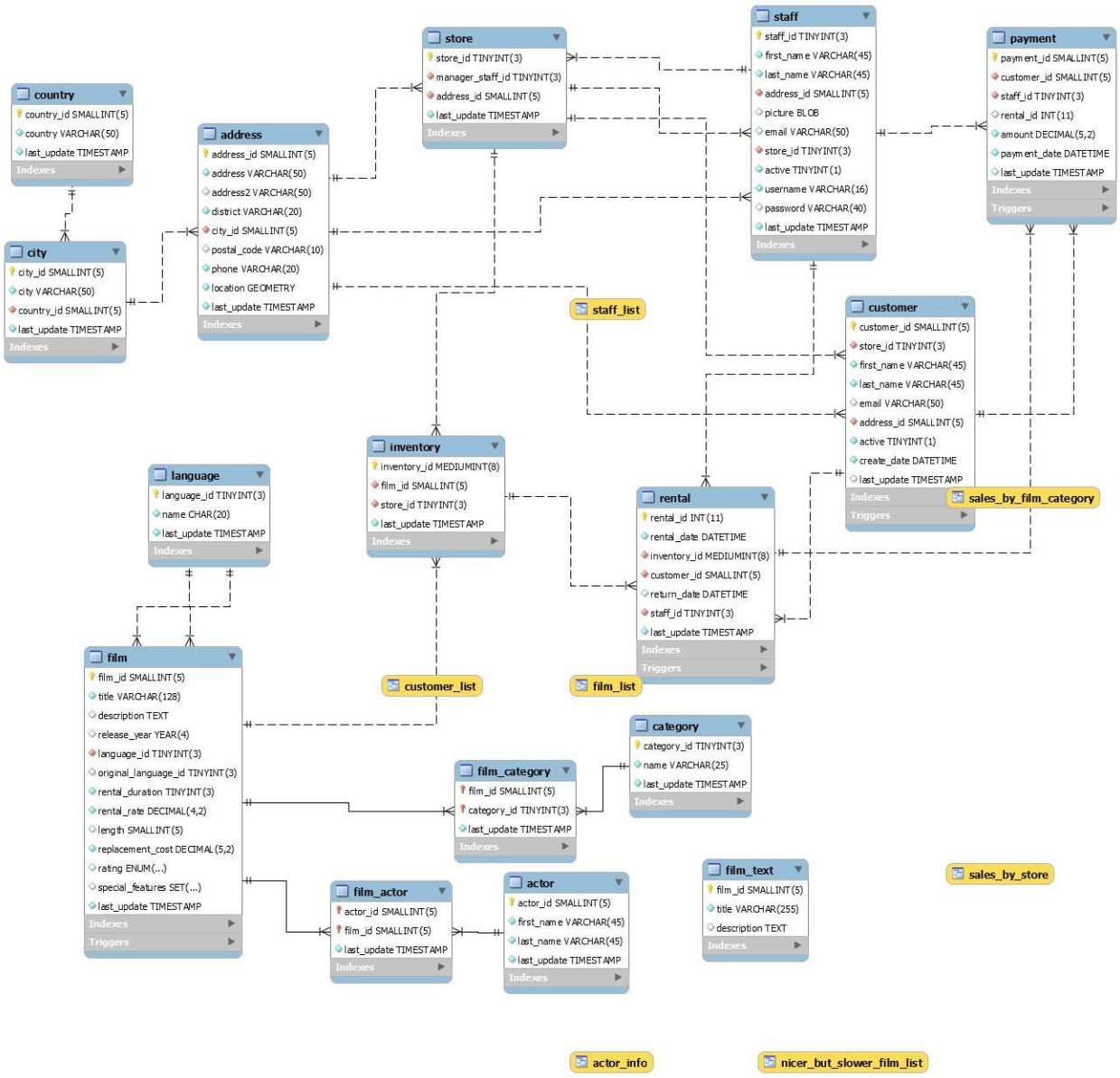
Project (due by Wednesday, March 5 at 11:59 PM).

Please make only one submission and put your full names into this Word document and submit this Word document. Please also submit a Tableau Workbook file .twb into the Final Project folder on Blackboard together with the Excel file of your Data Warehouse which you uploaded to Tableau and used for visualizations.

The script files sakila-schema.sql and sakila-data.sql create a database which contains tables (the database schema is below), with transactional data related to some company operations.

Please check early that you can create the sakila database on your machine. First, run this code - sakila-schema.sql, and second, run this code - sakila-data.sql.

Very Important! All three parts of the final project should be on One topic/subject of the data warehouse. For example, if you decide to track customers as your topic/subject, part 1 (Data Warehouse), part 2 (Queries) and part 3 (visualizations) should only be related to customers and should NOT include any other topics.



Please put all your work into **this single Word doc and also submit a Tableau Workbook file .twb and the Excel file of your Data Warehouse that you used for Tableau visualizations**. Please see instructions for Tableau below in question 3.

1. **(121 points)** Design and create a data warehouse for the provided database. The decisions about which fields to include and how to aggregate the data are left to you. You do not need to include every single data point from the tables given. Use your judgement as to what will be interesting/useful for the organization. But please make sure that you pull (combine) data from **at least six tables** and compute relevant aggregate statistics. Please compute relevant aggregate statistics for each table that you join. **In your queries later in part 2, you may join your Data Warehouse with other tables to answer useful questions**. Please see many examples from class lectures and you may adapt those codes for your purpose (for this dataset).

Submit a screenshot of the first 25 rows of your data warehouse (paste into this Word document) and the SQL code that you used to create it. Please copy and paste your SQL code into this Word document. If your PC does not show 25 rows of data, please submit what you have (i.e., rows you can see on a screenshot) with a comment that you cannot show 25 rows of data. Please add a full description of what your Data Warehouse will be tracking for a company. Please treat this assignment as a business case. So, the more you describe the better. Please also create an Excel file (Export from MySQL) of your data warehouse and use it for part 3 – Tableau visualizations.

The `film_category_performance_kpis` view provides a structured analysis of film rental data, focusing on store performance, customer engagement, and revenue trends. By aggregating information at the store, film category, and time (monthly and yearly) levels, this view enables businesses to make data-driven decisions regarding inventory, pricing, and customer retention.

This view combines data from multiple sources, including store details, manager information, rental transactions, customer data, and financial records. It links rental activity with specific film categories, allowing stakeholders to assess which genres perform best in different locations. Additionally, it includes key geographical details, such as store address, city, and country, to facilitate location-based analysis.

The primary KPIs measured include total rentals, total revenue, unique customers, and average rental duration. These indicators help assess rental demand, revenue contribution, and customer engagement with different film genres. The data is aggregated by year and month, making it possible to track trends over time and identify seasonal variations in rental behavior.
By leveraging this view, businesses can optimize store operations, improve customer satisfaction, and maximize revenue. It provides a comprehensive, data-driven foundation for strategic decision-making in the film rental industry.

```
CREATE OR REPLACE VIEW film_category_performance_kpis AS

SELECT
   s.store_id AS Store_ID,    st.first_name
AS Manager_First_Name,    st.last_name
AS Manager_Last_Name,    a.address AS
Store_Address,
```

```
   c.city AS Store_City,
co.country AS Store_Country,
cat.name AS Film_Category,
YEAR(r.rental_date) AS Year,
   MONTH(r.rental_date) AS Month,
   COUNT(r.rental_id) AS Total_Rentals,
   SUM(p.amount) AS Total_Revenue,
   COUNT(DISTINCT cu.customer_id) AS Total_Customers,
   AVG(DATEDIFF(r.return_date, r.rental_date)) AS Avg_Rental_Duration
FROM store s
JOIN staff st ON s.manager_staff_id = st.staff_id
JOIN address a ON s.address_id = a.address_id
JOIN city c ON a.city_id = c.city_id
JOIN country co ON c.country_id = co.country_id
JOIN inventory i ON s.store_id = i.store_id
JOIN rental r ON i.inventory_id = r.inventory_id
JOIN customer cu ON r.customer_id = cu.customer_id
JOIN payment p ON r.rental_id = p.rental_id
JOIN film f ON i.film_id = f.film_id
JOIN film_category fc ON f.film_id = fc.film_id
JOIN category cat ON fc.category_id = cat.category_id
WHERE r.rental_date IS NOT NULL
GROUP BY s.store_id, st.first_name, st.last_name, a.address, c.city, co.country,
cat.name, YEAR(r.rental_date), MONTH(r.rental_date); select * from
film_category_performance_kpis;
```

Result Grid Filter Rows: Search Export:

Store_ID	Manager_First_Name	Manager_Last_Na...	Store_Address	Store_City	Store_Country	Film_Category	Year	Month	Total_Rentals	Total_Revenue	Total_C
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Action	2005	5	48	222.52	45
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Action	2005	6	82	355.18	79
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Action	2005	7	252	946.48	209
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Action	2005	8	205	792.95	168
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Action	2006	2	9	24.91	9
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Animation	2005	5	41	169.59	39
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Animation	2005	6	86	350.14	78
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Animation	2005	7	239	963.61	196
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Animation	2005	8	190	775.10	156
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Animation	2006	2	12	38.85	12
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Children	2005	5	36	152.64	36
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Children	2005	6	61	245.39	59
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Children	2005	7	200	768.00	173
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Children	2005	8	152	580.48	136
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Children	2006	2	4	11.96	4
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Classics	2005	5	29	106.71	28
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Classics	2005	6	64	246.36	60
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Classics	2005	7	178	666.22	159
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Classics	2005	8	172	682.28	149
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Classics	2006	2	4	15.94	4
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Comedy	2005	5	40	160.60	38
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Comedy	2005	6	72	364.28	67
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Comedy	2005	7	206	1011.94	174
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Comedy	2005	8	177	821.23	157
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Comedy	2006	2	7	19.92	7
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Documentary	2005	5	35	152.65	34
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Documentary	2005	6	64	249.36	63
1	Mike	Hillyer	47 MySakila Drive	Lethbridge	Canada	Documentary	2005	7	180	751.20	144

2. **(104 points)** Create **eight** SQL queries **on your data warehouse** (not on the original dataset) that answer interesting questions. At least **6** queries should be more complex queries. For example, more complex queries could include Joins, a Group By, UNION elements or a subquery or use some aggregate functions and summary calculations and conditional logic codes (see examples in the class lectures' slides). **If needed, you may join your Data Warehouse with other tables (which are not a part of Data Warehouse) to answer useful questions.**

Submit a copy of each query SQL code (paste into this Word document), and the screenshot of each query results (or the first 25 rows if it is longer or how many rows you can get on your PC) and full description of the question your SQL code was addressing and what you found in the results. The question that each query answers should be useful for a company to make decisions and act upon.

1. Top Performing Film Categories

```
SELECT cat.name AS category,
    COUNT(r.rental_id) AS total_rentals,
    SUM(p.amount) AS total_revenue
FROM rental r
JOIN payment p ON r.rental_id = p.rental_id
JOIN inventory i ON r.inventory_id = i.inventory_id
JOIN film f ON i.film_id = f.film_id
JOIN film_category fc ON f.film_id = fc.film_id
JOIN category cat ON fc.category_id = cat.category_id
GROUP BY cat.name
ORDER BY total_revenue DESC
LIMIT 10;
```

category	total_rentals	total_revenue
Sports	1179	5314.21
Sci-Fi	1101	4756.98
Animation	1166	4656.30
Drama	1060	4587.39
Comedy	941	4383.58
Action	1112	4375.85
New	940	4351.62
Games	969	4281.33
Foreign	1033	4270.67
Family	1096	4226.07

2. Customer Lifetime Value (CLV) Analysis

```
SELECT c.customer_id,
    CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
    COUNT(r.rental_id) AS total_rentals,
    SUM(p.amount) AS total_spent
FROM customer c
JOIN rental r ON c.customer_id = r.customer_id
JOIN payment p ON r.rental_id = p.rental_id
GROUP BY c.customer_id, customer_name
ORDER BY total_spent DESC
LIMIT 10;
```

customer_id	customer_name	total_rentals	total_spent
526	KARL SEAL	45	221.55
148	ELEANOR HUNT	46	216.54
144	CLARA SHAW	42	195.58
137	RHONDA KENNEDY	39	194.61
178	MARION SNYDER	39	194.61
459	TOMMY COLLAZO	38	186.62
469	WESLEY BULL	40	177.60
468	TIM CARY	39	175.61
236	MARCIA DEAN	42	175.58
181	ANA BRADLEY	34	174.66

3. Store Performance by Year

```
SELECT Store_ID, Year,
    SUM(Total_Revenue) AS Revenue,
    SUM(Total_Customers) AS Unique_Customers
FROM film_category_performance_kpis
GROUP BY Store_ID, Year
ORDER BY Revenue DESC;
```

Store_ID	Year	Revenue	Unique_Customers
2	2005	33455.69	6985
1	2005	33436.69	6813
2	2006	271.08	90
1	2006	243.10	90

4. Top Performing Actors

```
SELECT a.actor_id,
    CONCAT(a.first_name, ' ', a.last_name) AS actor_name,
    SUM(p.amount) AS total_revenue
FROM film_actor fa
JOIN actor a ON fa.actor_id = a.actor_id
JOIN film f ON fa.film_id = f.film_id
JOIN inventory i ON f.film_id = i.film_id
JOIN rental r ON i.inventory_id = r.inventory_id
JOIN payment p ON r.rental_id = p.rental_id
GROUP BY a.actor_id, actor_name
ORDER BY total_revenue DESC
LIMIT 10;
```

actor_id	actor_name	total_revenue
107	GINA DEGENERES	3442.49
181	MATTHEW CARREY	2742.19
198	MARY KEITEL	2689.25
81	SCARLETT DAMON	2655.28
102	WALTER TORN	2620.62
144	ANGELA WITHERSPOON	2614.46
58	CHRISTIAN AKROYD	2611.49
60	HENRY BERRY	2602.88
28	WOODY HOFFMAN	2546.40
111	CAMERON ZELLWEGER	2529.41

5. Most rented films

```
SELECT f.title AS Film_Title,
    COUNT(r.rental_id) AS Total_Rentals
FROM film f
JOIN inventory i ON f.film_id = i.film_id
JOIN rental r ON i.inventory_id = r.inventory_id
GROUP BY f.title
ORDER BY Total_Rentals DESC
LIMIT 10;
```

Film_Title	Total_Rentals
BUCKET BROTHERHOOD	34
ROCKETEER MOTHER	33
RIDGEMONT SUBMARINE	32
GRIT CLOCKWORK	32
SCALAWAG DUCK	32
JUGGLER HARDLY	32
FORWARD TEMPLE	32
HOBBIT ALIEN	31
ROBBERS JOON	31
ZORRO ARK	31

6. Revenue by Film Length

```
SELECT CASE
        WHEN f.length <= 60 THEN 'Short (<60 min)'
        WHEN f.length BETWEEN 61 AND 120 THEN 'Medium (61-120 min)'
        ELSE 'Long (>120 min)'
    END AS Film_Length_Category,
    COUNT(r.rental_id) AS Total_Rentals,
    SUM(p.amount) AS Total_Revenue
FROM film f
JOIN inventory i ON f.film_id = i.film_id
JOIN rental r ON i.inventory_id = r.inventory_id
JOIN payment p ON r.rental_id = p.rental_id
GROUP BY Film_Length_Category
ORDER BY Total_Revenue DESC;
```

Film_Length_Category	Total_Rentals	Total_Revenue
Long (>120 min)	7300	30868.01
Medium (61-120 min)	7093	29864.07
Short (<60 min)	1651	6674.48

7.Store Level Film Performance

```
SELECT s.store_id AS Store_ID,
c.name AS Film_Category,
SUM(p.amount) AS Total_Revenue,
    COUNT(r.rental_id) AS Total_Rentals
FROM store s
JOIN inventory i ON s.store_id = i.store_id
JOIN rental r ON i.inventory_id = r.inventory_id
JOIN payment p ON r.rental_id = p.rental_id
JOIN film f ON i.film_id = f.film_id
JOIN film_category fc ON f.film_id = fc.film_id
JOIN category c ON fc.category_id = c.category_id
GROUP BY s.store_id, c.name
ORDER BY Total_Revenue DESC
Limit 10;
```

Store_ID	Film_Category	Total_Revenue	Total_Rentals
2	Sports	2825.75	624
1	Drama	2573.24	576
2	Sci-Fi	2553.19	580
1	Sports	2488.46	555
1	New	2402.98	502
1	Comedy	2377.97	502
2	Animation	2359.01	598
2	Documentary	2348.01	600
1	Action	2342.04	596
1	Animation	2297.29	568

8. Top Grossing Film

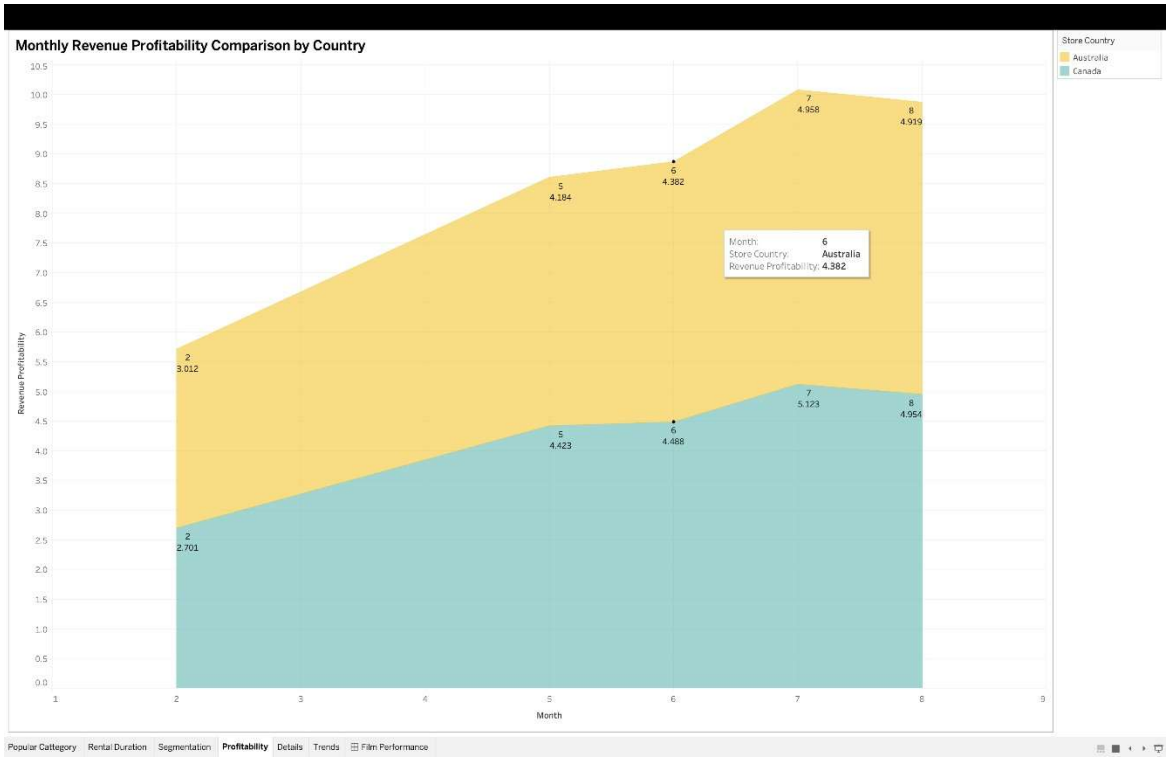
```
SELECT f.title AS Film_Title,
      SUM(p.amount) AS Total_Revenue,
      COUNT(r.rental_id) AS Total_Rentals
FROM film f
JOIN inventory i ON f.film_id = i.film_id
JOIN rental r ON i.inventory_id = r.inventory_id
JOIN payment p ON r.rental_id = p.rental_id
GROUP BY f.title
ORDER BY Total_Revenue DESC
LIMIT 10;
```

Film_Title	Total_Revenue	Total_Rentals
TELEGRAPH VOYAGE	231.73	27
WIFE TURN	223.69	31
ZORRO ARK	214.69	31
GOODFELLAS SALUTE	209.69	31
SATURDAY LAMBS	204.72	28
TITANS JERK	201.71	29
TORQUE BOUND	198.72	27
HARRY IDAHO	195.70	30
INNOCENT USUAL	191.74	26
HUSTLER PARTY	190.78	22

3. (**75 points**) Create **five** Tableau individual visualizations (graphs) **on your data warehouse** (**plus one dashboard** as discussed below) with valuable information to present findings to senior management of the company. Save each visualization as a png file (as I show in class, and we will also practice in the lab 5) and paste each individual visualization png file **into this Word** document with the full explanation of what the visualizations show, how they are useful to a company and how company management could make decisions based on what you show. Finally, combine those **five** visualizations into one **Dashboard** (as I show in class, and we will also practice in the lab 5), and save this Dashboard as a png file and **paste the Dashboard into this Word** document.

Important! You might visualize some queries from part 2 or create completely new visualizations. If for queries (in part 2), you used data warehouse together with (joining) some additional tables, you need to create a new source Excel file for Tableau, and submit it as described below in the next paragraph (please explain in your submission what source you used). Alternatively, you may do visualizations only on Data Warehouse Excel file (from part 1) and submit that file as a source file for Tableau (see below). In general, we need one Excel source file for Tableau to check your visualizations and the Tableau workbook (see below).

Please also save the whole Tableau project as a Tableau Workbook file .twb (In Tableau use File - Save as) and submit to the Final Project folder on Blackboard together with this Word document and together with the Excel file of your Data Warehouse which you uploaded to Tableau and used for visualizations.



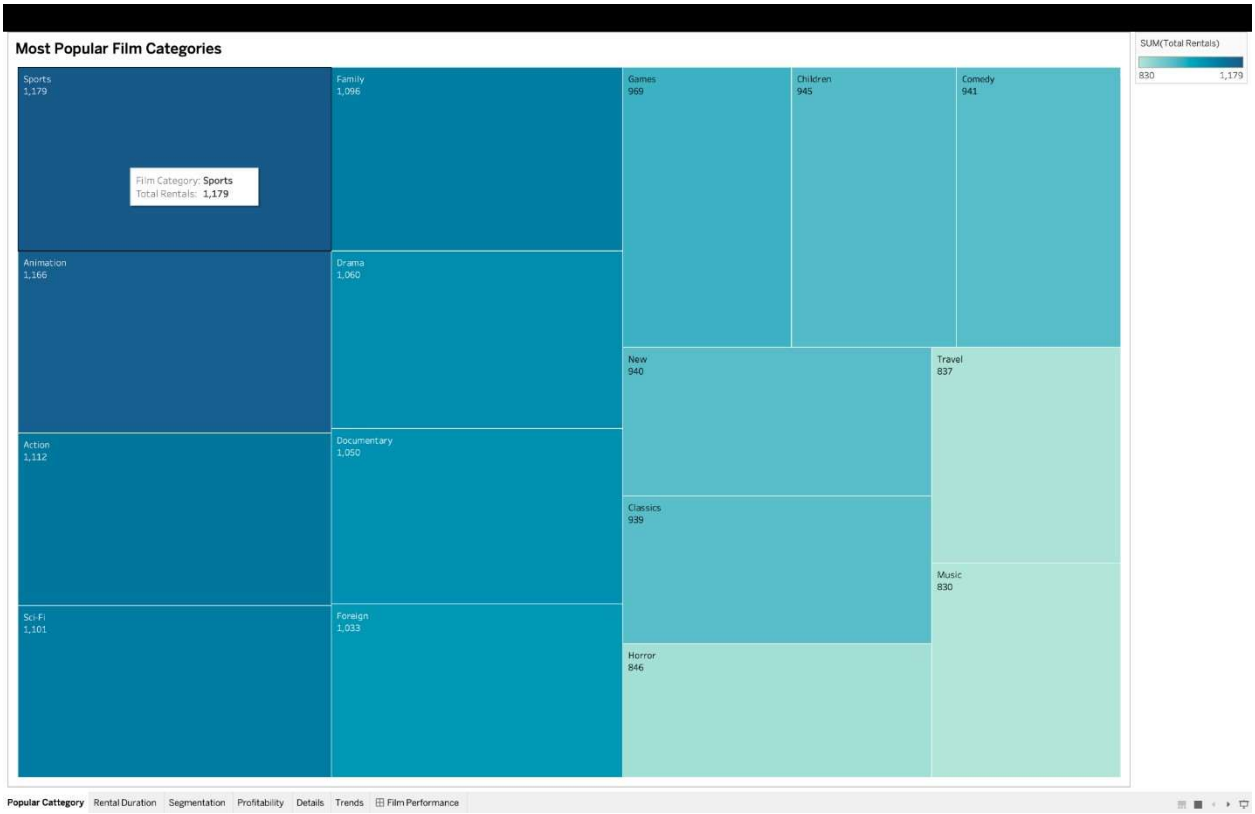
Monthly Revenue Profitability Comparison by Country:

Explanation:

This visualization highlights film rental profitability trends across Australia and Canada over several months. Australia consistently generates higher profitability, indicating a stronger rental market or better-performing film selections. Both countries show growth, but Australia experiences a slight dip in the last month, suggesting potential changes in demand or pricing effectiveness.

Recommendation:

To enhance film performance and profitability, the company should analyze top-performing film genres and customer preferences in Australia and apply those insights to Canada. Additionally, the decline in Australia warrants a closer look at rental pricing strategies or seasonal demand fluctuations. Adjusting film inventory and promotional strategies based on these insights can maximize profitability across both markets**.**

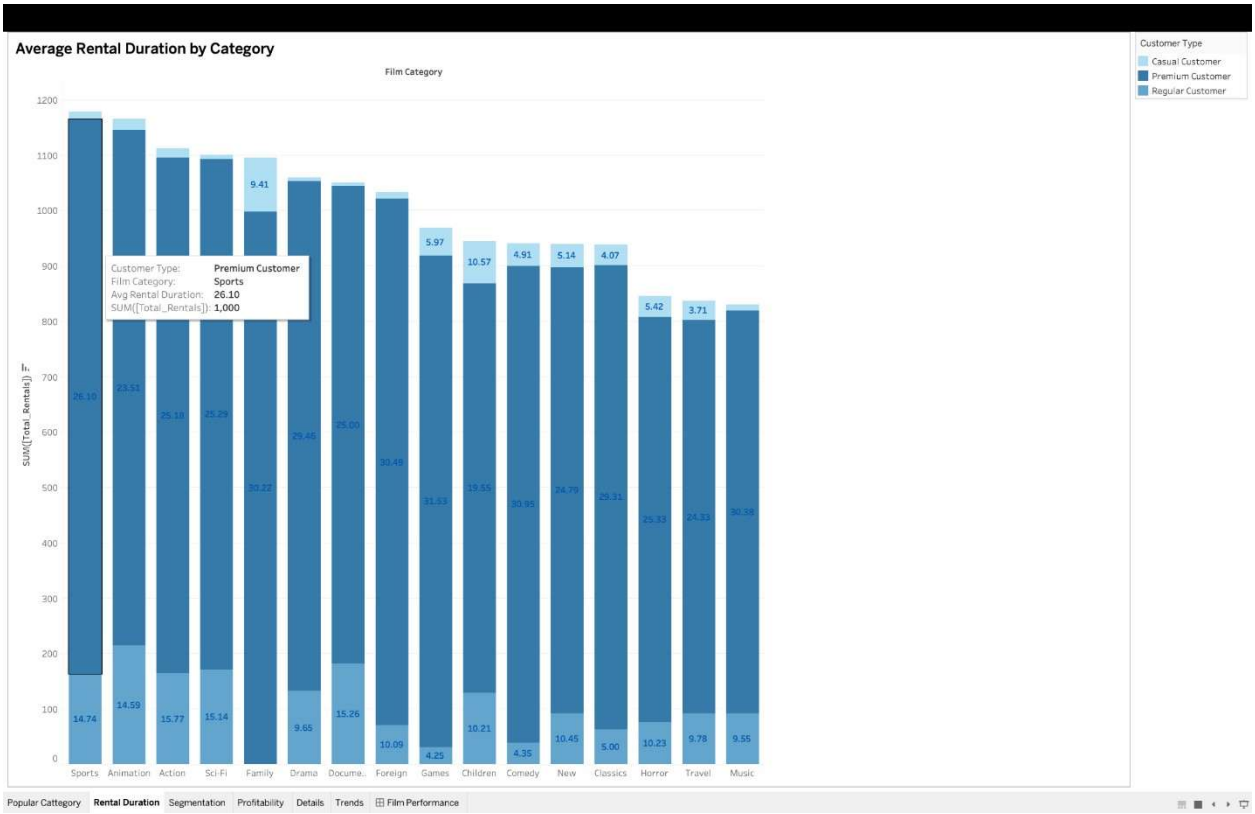


Most Popular Film Categories Explanation:

This treemap visualization showcases the most popular film categories based on total rentals. Sports, Animation, and Action films lead in popularity, while categories like Music and Horror have relatively lower rentals. The varying sizes of the blocks indicate the rental volume distribution across different genres.

Recommendation:

To maximize film performance and profitability, the company should increase inventory and marketing efforts for top-performing categories, such as Sports and Animation, to sustain demand. For low-performing categories, consider targeted promotions, bundling strategies, or evaluating whether to reduce inventory and reallocate resources to higher-demand genres.

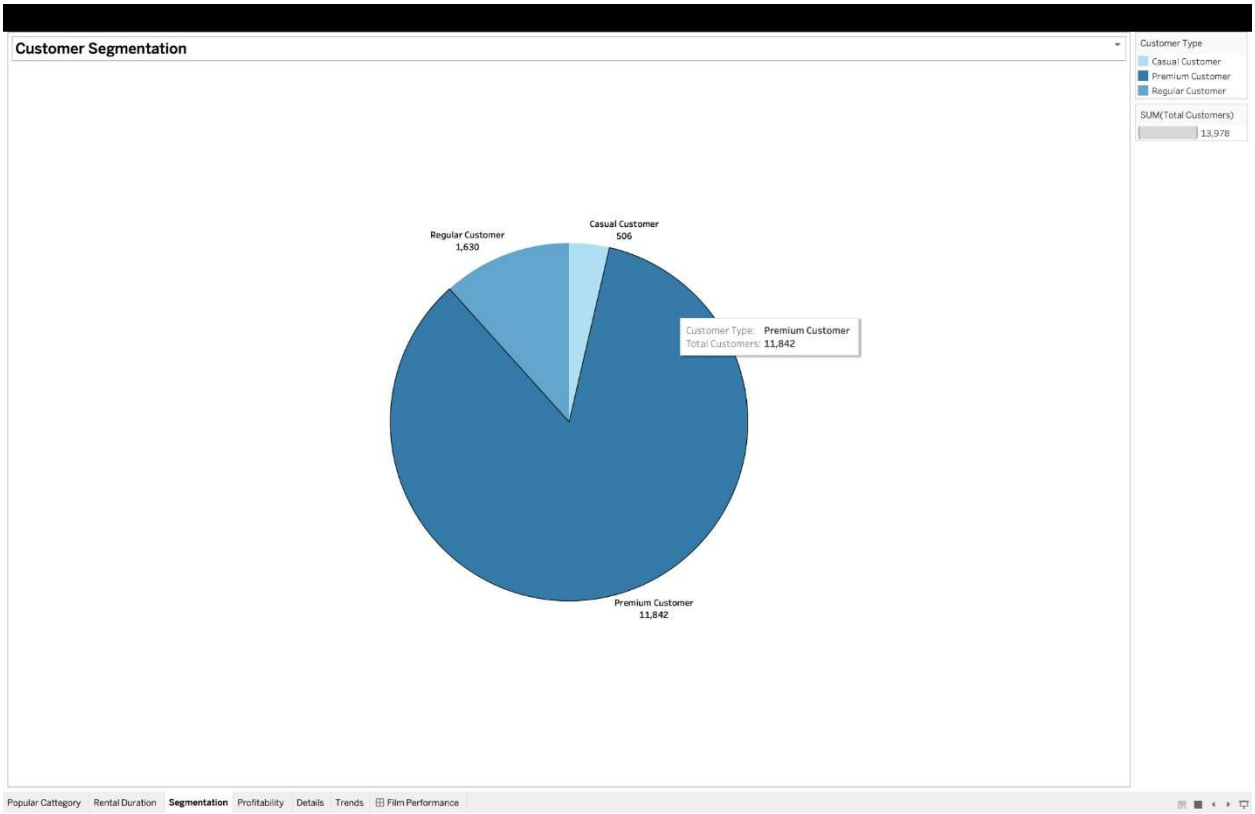


Average Rental Duration by Category Explanation:

This bar chart visualizes the average rental duration across different film categories, segmented by customer type (Premium, Regular, and Casual Customers). Sports, Drama, and Foreign films have the longest rental durations, while categories like Games, Comedy, and Horror have shorter rental periods. Premium customers tend to rent films for the longest duration, whereas casual customers have significantly shorter rental periods.

Recommendation:

To optimize film inventory and rental strategies, the company should consider adjusting rental pricing based on duration—for instance, charging premium customers higher fees for extended rentals. Additionally, shorter-duration films (e.g., Games and Comedy) could benefit from special rental promotions to boost engagement and profitability. Understanding these trends allows for better inventory management and targeted marketing strategies based on customer rental behavior.

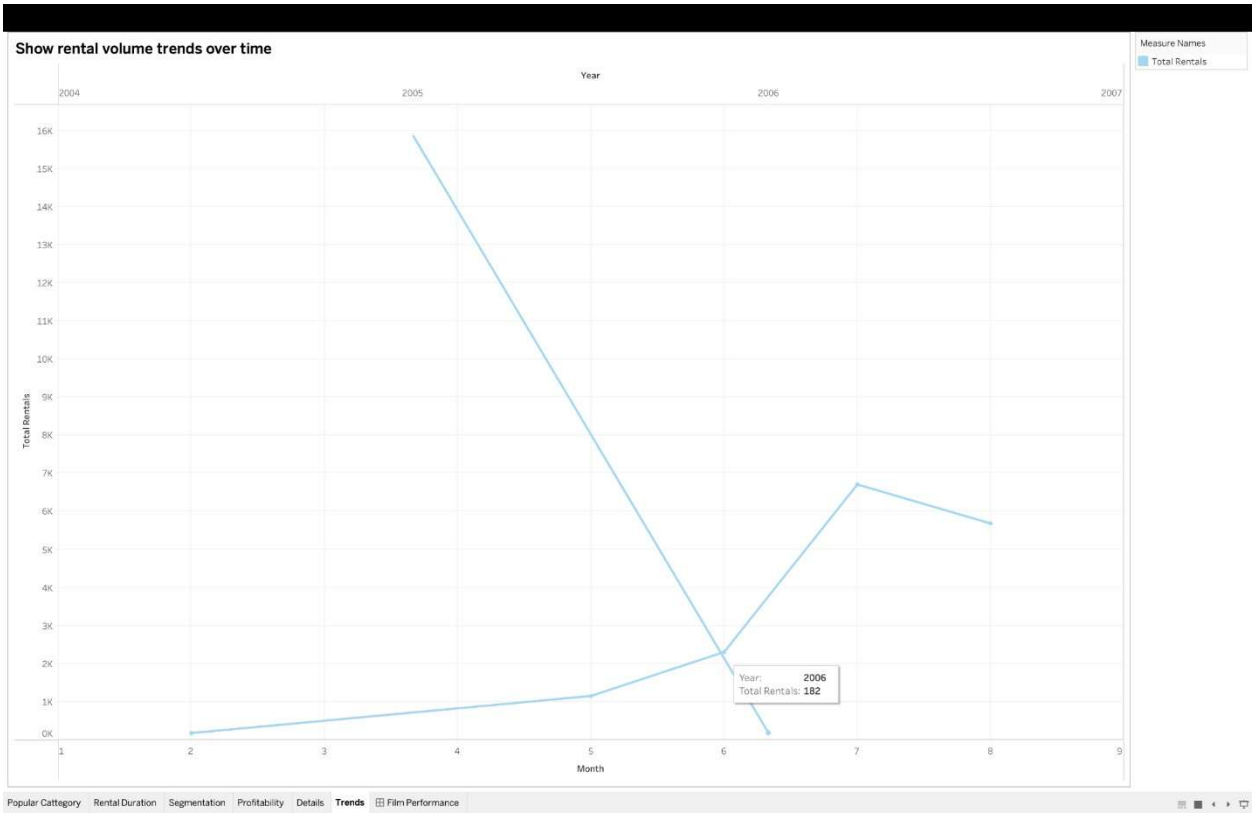


Customer Segmentation Explanation:

This pie chart categorizes customers into Premium, Regular, and Casual segments, with Premium Customers making up the majority. Regular and Casual customers form a much smaller portion of the total customer base, indicating a strong reliance on loyal, high-value customers.

Recommendation:

To enhance film performance and revenue, the company should focus on retaining and expanding the Premium segment through exclusive promotions, early access to new films, or loyalty rewards. Simultaneously, efforts should be made to convert Regular and Casual customers into Premium members through targeted offers, discounts on longer rentals, or subscription-based incentives.

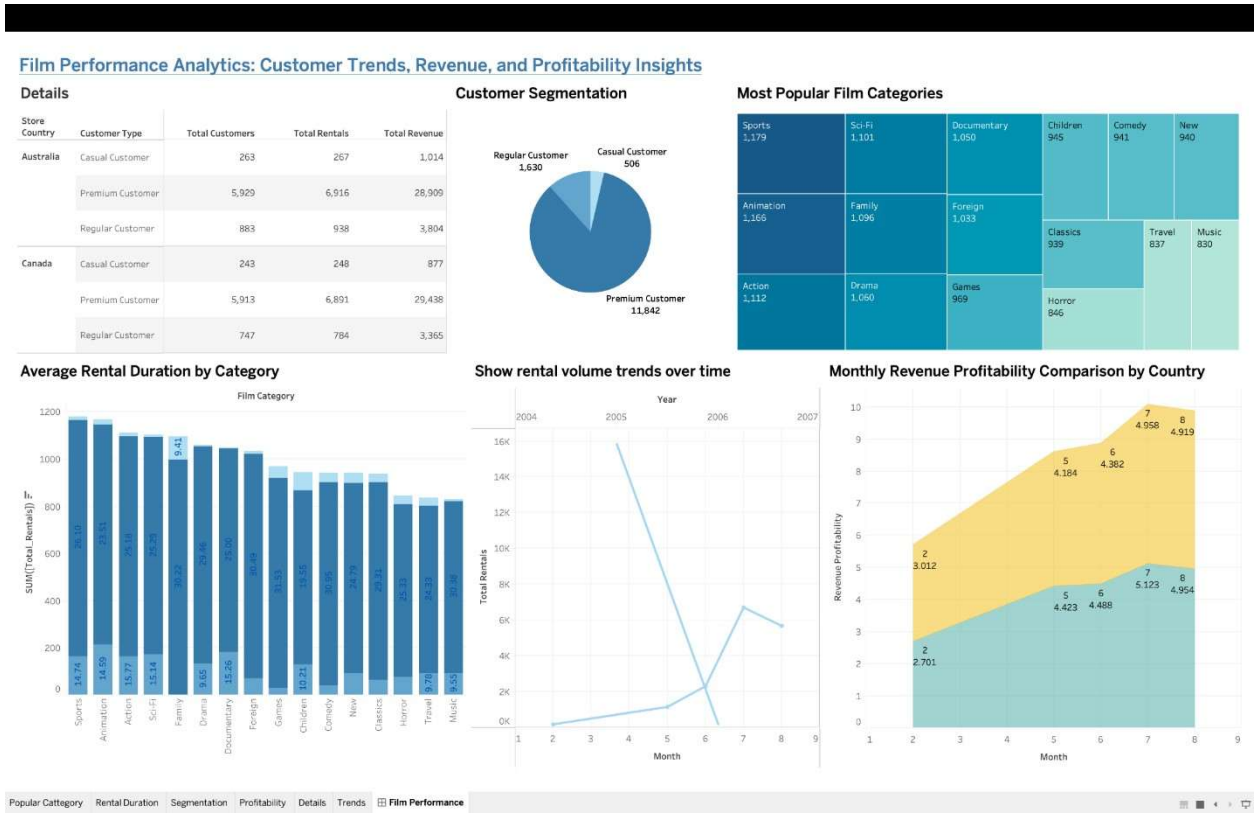


Rental Volume Trends Over Time Explanation:

This line chart tracks total film rental volumes over time, revealing significant fluctuations. The steep drop after 2005 suggests a decline in rental activity, followed by some recovery in 2006. Such variations may be due to market shifts, pricing changes, seasonality, or competitive factors affecting customer behavior.

Recommendation:

To stabilize and grow rental volumes, the company should analyze the reasons behind the 2005 decline—whether due to pricing, customer preferences, or external competition. Introducing promotional campaigns, loyalty incentives, or new film selections can help regain and sustain customer engagement. Additionally, identifying peak rental periods can aid in better inventory planning and marketing strategies to boost long-term performance.



Store Country	Customer Type	Total Customers	Total Rentals	Total Revenue
Australia	Casual Customer	263	267	1,014
	Premium Customer	5,929	6,916	28,909
	Regular Customer	883	938	3,804
Canada	Casual Customer	243	248	877
	Premium Customer	5,913	6,891	29,438
	Regular Customer	747	784	3,365

The Film Performance Analytics Dashboard provides key insights into customer behavior, film popularity, and rental profitability. Premium customers dominate the market, contributing the highest revenue, while casual and regular customers represent a smaller portion, highlighting the need for targeted engagement strategies.

Sports, Animation, and Action films are the most rented categories, indicating strong demand. Rental duration varies by genre, with premium customers keeping films longer, particularly in Sports and Drama. Optimizing pricing and promotions based on these trends can enhance profitability.

Rental trends over time show a decline after 2005, suggesting the need for strategic interventions such as promotional campaigns or content adjustments. Profitability comparison between Australia and Canada highlights Australia's stronger performance, signaling an opportunity to replicate successful strategies in Canada to boost rentals and revenue.

This dashboard serves as a strategic tool for improving customer retention, inventory management, and overall business growth.

General grading criteria: Your completed work will be evaluated using the criteria below. I encourage you to use your creativity and other business skills (communication,

presentation, critical thinking) in addition to the data management concepts and the SQL and Tableau skills that we have covered in CIS467.

High score	**Score between high and good**	**Good/medium score**	**Low score**
All required parts of the final project are complete and technically correct. Queries are useful/interesting and provide valuable information for senior management to act upon. Not just random queries. Tableau visualizations provide interesting useful information based on which senior management of the company can make important decisions.	All required parts of the final project are complete and technically correct (with possibly a few minor errors). Queries are useful/interesting and provide valuable information for senior management to act upon. Not just random queries (with possibly a few minor errors). Tableau visualizations provide interesting useful information based on which senior management of the company can make important decisions (with possibly a few minor errors).	Some required parts of the final project are missing and/or there are more significant errors. Some queries appear random and do not answer any useful/interesting questions. Tableau visualizations are very simple but may still provide interesting useful information based on which senior management of the company can make important decisions.	The final project has large portions missing and/or major conceptual errors. Most/all queries (if any) appear random and do not answer any useful/interesting questions. Tableau visualizations are very simple and **do not** provide interesting useful information based on which senior management of the company can make important decisions.